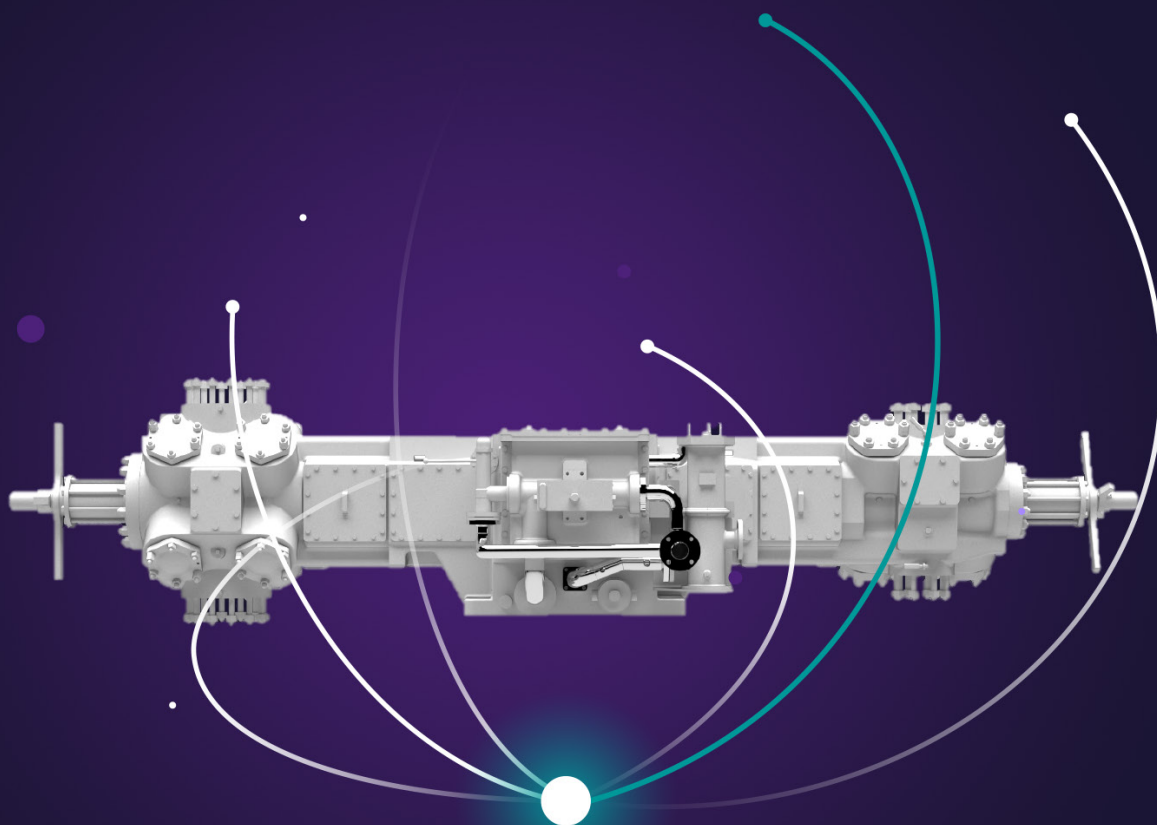


HOSS standard materials of construction



Frame and running gear

Component	Material	Industry standard
Capscrews	Carbon steel	ASTM A449
Connecting rod	Alloy steel	AISI 4142
Connecting rod bushing	Bronze babbitt	SAE 64, SAE 19
Crankpin bearing	Tri-metal	SAE 1020, SAE 19, SAE 49
Crankshaft (5" stroke)	Forged steel	AISI 4045
Crankshaft (6" stroke)	Forged steel	AISI 4140
Crankshaft (7" stroke)	Forged steel	AISI 4140
Crankshaft oil ring	Gray iron	ASTM A48
Crosshead	Nodular iron	ASTM A536
Crosshead pin	Alloy steel	AISI 4140
Crosshead shoe	Aluminum	ASTM B26
Crosshead shoe bolt	Alloy steel	ASTM A574
Distance piece	Gray iron	ASTM A48
Frame	Gray iron	ASTM A48
Frame covers	Carbon steel	ASTM A283
Frame drive end cover	Gray iron	ASTM A48
Frame pump end cover	Gray iron	ASTM A48
Frame tie rod	Alloy steel	ASTM A193
Frame tie rod spacer	Gray iron	ASTM A48
Main bearing cap	Gray iron	ASTM A48
Main bearings	Tri-metal	SAE 1020, SAE 19, SAE 49
Nuts	Carbon steel	ASTM A194
Studs	Alloy steel	ASTM A193

Cylinder gas end

Component	Material	Industry standard
Capscrews	Carbon steel	ASTM A449
Combination piston/rider rings	Carbon filled ptfe	-----
Cover valve	Nodular iron, steel	ASTM A536, AISI 4140
Cylinder casting	Nodular iron	ASTM A536
Cylinder forging	Forged steel	AISI 4132
Nuts	Carbon steel	ASTM A194
Outer head	Nodular iron	ASTM A536
Packing case	Cast iron, steel	ASTM A278, AISI 4140
Packing pressure rings	Carbon filled ptfe, cast iron	-----
Piston rod	Alloy steel	AISI 4140
Piston: Ø6.00" – Ø10.50"	Gray iron	ASTM A48
Piston: Ø11.00" – Ø14.00"	Gray cast iron	ASTM A278
Piston: Ø15.00" – Ø28.00"	Aluminium alloy, anodized	ASTM B26
Studs	Alloy steel	ASTM A193
Valve plates & elements	"Hi temp" peek (non-metallic)	-----
Valve seat & guard	Nodular iron	ASTM A536

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